Military Institute of Science and Technology
Department of Computer Science and Engineering
Course: Data Structure and Algorithm-II Sessional
Course Code: CSE-216
Assignment
Topic: String Matching Algorithm (KMP)

A string is a finite sequence of symbols that are chosen from an alphabet. In this problem you are given two non-empty strings A and B, both contain lower case English alphabets. You have to find one, several, or all occurrences of a defined string (B) in a large string (A) such that each matching is perfect. All alphabets of B must be matched to corresponding matched subsequence. Additionally to find the number of times B occurs as a substring of A.

Input

Input starts with an integer **T (≤ 5),** denoting the number of test cases.

Each case starts with two lines. First line contains A and second line contains B. You can assume than **1 ≤ length(A), length(B) ≤ 10^6.**

Output

For each case, print the case number and the staring indices of **B** if found in **A** along with the number of times **B** occurs as a substring of **A**.

Sample Input	Sample Output
4 axbyczd abc abcabcabcabc abc aabacbaabbaaz aab aaaaaa aa	Case 1: Not Found 0 times Case 2: Found at 0, 3, 6, 9 position 4 times Case 3: Found at 0, 6 position 2 times Case 4: Found at 0, 1, 2, 3, 4 5 times